Nishal Stanislaus Silva, *PhD*

Researcher, Real-Time Edge ML Systems | Edge-to-Server Computing for Audio, Sensor and Time-Series Data

Toronto, ON, Canada | nishal.silva@hotmail.com | +1 613 200 2030

nishal.xyz | Google Scholar | ORCID 0000-0003-0406-7459 | github.com/ns2max

Canadian Permanent Resident, eligible to work in Canada without sponsorship

Research Interests

Edge-to-server computing for real-time machine learning: partitioning latency-critical inference between embedded devices and servers, latency and throughput characterization, and benchmark design; low-latency ML on embedded and edge hardware; time-series and sequential pattern detection; audio signal processing and music information retrieval; multimodal sensing (audio, IMU, EEG, mixed-reality telemetry); smart musical instruments and the Internet of Musical Things; benchmark and dataset design; human-centred evaluation of ML-driven interactive systems.

Education

Ph.D., Information Engineering and Computer Science 2021 – Jan 2025
University of Trento, Italy (DISI; CIMIL lab). Supervisor: Prof. Luca Turchet

- Thesis: *Embedded Real-time Musical Pattern Detection for Smart Musical Instruments* (defended 31 Jan 2025). iris.unitn.it/handle/11572/445071

M.Sc., Telecommunication and Electronic Engineering Aug 2018
Sheffield Hallam University, UK

- Thesis: *On Musical Onset Detection via the S-Transform*; published at Asilomar 2018 and arXiv:1712.02567.

B.Eng. (Hons), Electronic Engineering Mar 2014
Sheffield Hallam University, UK

- Thesis: *A Hand Gesture Controlled TV Remote*.

Academic Appointments

Postdoctoral Researcher Jan 2025 – Dec 2025
CIMIL, University of Trento, Italy | EU Horizon EIC Pathfinder project MUSMET (grant 101184379)

- Owned the full ML pipeline for streaming audio and IMU/gesture data: acquisition, DSP and feature engineering, training and evaluation, edge deployment (Raspberry Pi 4, Elk Audio OS, VST, OSC) and monitoring.
- Published system detects audio patterns in 14 ms on a Raspberry Pi 4 (F1 0.76) against 1038 ms for a DTW baseline; real-time inference under 30 ms on edge-class devices.
- Designed benchmark suites (baselines and ablations: RNN vs. DTW, audio vs. MIDI, pattern-set sizes 1/3/10) to quantify accuracy, latency and CPU trade-offs.
- Built a real-time multimodal extraction and synchronization pipeline (audio, EEG/BCI, head and hand tracking from mixed-reality headsets) across four musicians simultaneously, with SAE Institute Barcelona and g.tec medical engineering.
- Co-designed and ran two live multisensory concerts (20 audience members, 6 performers) used as a user study.
- Supervised two BSc theses (see Supervision). Four peer-reviewed outputs in 2025.

Visiting Researcher May 2024 – Aug 2024
IDMIL / CIRMMT, McGill University, Montreal, Canada. Host: Prof. Marcelo M. Wanderley

- Built a self-powered smart electric guitar (BNO055 IMU, HiFiBerry, Raspberry Pi 4) with real-time gesture detection (LSTM) fused with audio pattern detection through a hierarchical finite state machine; F1 0.78 on a 500-event corpus.

- Case study with five professional guitarists; compute and power profiling for edge feasibility; explored diffusion- and VAE-based synthetic data for low-label regimes.

Visiting Researcher Jan 2024 – Mar 2024
University of Visual and Performing Arts, Colombo, Sri Lanka

- Ran a user study of musicians' perception and use of smart musical instruments with embedded pattern detection (published at I3DA 2025).

Peer-Reviewed Publications

Journal articles

[1] **N. Silva** and L. Turchet. Real-Time Audio Pattern Detection for Smart Musical Instruments. *Journal of the Audio Engineering Society*, 74(3), Mar 2026. DOI 10.17743/jaes.2022.0250.

[2] **N. Silva** and L. Turchet. Smart Drums: Comparing Audio and MIDI in Embedded Real-Time Drum Pattern Recognition. *Journal of the Audio Engineering Society*, in press (2026).

Conference papers

[3] **N. Silva** and L. Turchet. Music Information Retrieval as a Service (MIRaaS): Latency Characterization of an Edge-to-Server Architecture for Near-Real-Time Audio Analysis. *7th IEEE International Symposium on the Internet of Sounds (IS² 2026)*, accepted.

[4] **N. Silva**, M. Wanderley and L. Turchet. Melody and Motion: Integrating Guitar Gestures with Musical Patterns for Extended Control in Live Performance and Metaverse Applications. *IEEE 6th International Symposium on the Internet of Sounds (IS² 2025)*, pp. 1–9, Oct 2025. DOI 10.1109/IS264627.2025.11284634.

[5] **N. Silva**, A. Boem and L. Turchet. Interactive IoMusT-Based Concerts: Real-Time Pattern Recognition and Audience Experience. *IEEE International Conference on Immersive and 3D Audio (I3DA 2025)*, Sep 2025. DOI 10.1109/i3da65421.2025.11202042.

[6] **N. Silva** and L. Turchet. User-Centered Evaluation of Smart Musical Instruments with Embedded Real-Time Pattern Detection. *IEEE I3DA 2025*, Sep 2025. DOI 10.1109/i3da65421.2025.11202088.

[7] **N. Silva** and L. Turchet. Real-Time Pattern Recognition of Symbolic Monophonic Music. *Audio Mostly 2024*, ACM, Sep 2024. DOI 10.1145/3678299.3678329.

[8] **N. Silva**. Towards an Internet of Musical Things (IoMusT) Ecosystem: Multi-Sensory Performances Enabled Through AI-Based Smart Musical Instruments. *11th International Conference on Arts and Humanities (ICOAH)*, Mar 2024.

[9] **N. Silva** and L. Turchet. A Structural Similarity Index Based Method to Detect Symbolic Monophonic Patterns in Real-Time. *25th International Conference on Digital Audio Effects (DAFx20in22)*, Vienna, Sep 2022.

[10] **N. Silva**, C. Fischione and L. Turchet. Towards Real-Time Detection of Symbolic Musical Patterns: Probabilistic vs. Deterministic Methods. *27th Conference of Open Innovations Association (FRUCT)*, pp. 238–246, 2020.

[11] **N. Silva**, P. C. Weeraddana and C. Fischione. On Musical Onset Detection via the S-Transform. *52nd Asilomar Conference on Signals, Systems and Computers*, pp. 1080–1085, Oct 2018. DOI 10.1109/ACSSC.2018.8645367.

Demo papers

[12] **N. Silva** and L. Turchet. Demo of a Smart Musical Instrument-Based Real Time Pattern Detection System. *Audio Mostly 2024*, ACM, pp. 552–554. DOI 10.1145/3678299.3678359.

Theses and preprints

[13] **N. Silva**. Embedded Real-time Musical Pattern Detection for Smart Musical Instruments. Ph.D. thesis, University of Trento, 2025.

[14] **N. Silva** and P. C. Weeraddana. On Musical Onset Detection via the S-Transform. arXiv:1712.02567, Dec 2017.

Google Scholar: 27 citations, h-index 3. Full list: nishal.xyz/publications.

Open Datasets and Software

Datasets (Zenodo, open access; 9,839 files from 70+ musicians in total)

- **DoMP**, Dataset of Monophonic Patterns: 4,392 MIDI files, 40 performers (piano, guitar). DOI 10.5281/zenodo.10818617.
- **DoPP**, Dataset of Polyphonic Patterns: 2,276 audio recordings (44.1 kHz, 24-bit), 20 performers, 176 patterns. DOI 10.5281/zenodo.14497998.
- **DoDP** and **DoDP2**, Datasets of Drum Patterns: 994 and 2,177 MIDI files, 10 drummers each. DOIs 10.5281/zenodo.14497974 and 10.5281/zenodo.18395007.

Software

- **Nebula**: open-source C++17 library for audio feature extraction with per-feature latency benchmarks on ARM and Raspberry Pi. github.com/ns2max/nebula
- **MIRaaS**: edge-to-server Music Information Retrieval service that offloads analysis from embedded devices to a server over UDP, characterized for latency (IS² 2026).
- **VarianceEngine** (in progress): LoRA fine-tuning of MusicGen-medium with a custom audio-to-audio conditioning head, trained across two GPUs (RTX 4090 and RTX 3090).
- **LiveLaTeX**: VS Code extension for real-time LaTeX compile and preview (Marketplace: `ns2max.livelatex`).

Grants and Projects

No principal-investigator funding to date (postdoctoral researcher on a funded consortium project). No career interruptions to declare. Tier 2 CRC: no prior independent academic appointment.

MUSMET, Musical Metaverse made in Europe 2025

EU Horizon EIC Pathfinder Open, grant 101184379, 3.0 M EUR, coordinated by University of Trento (12 partners)

- Postdoctoral researcher (not principal investigator); responsible for real-time pattern detection, edge deployment and the multimodal data pipeline.

Awards and Recognition

- **Finalist**, MIDI Innovation Awards 2023, Prototypes and Non-Commercial Software ("Hot Licks"; 15 finalists worldwide; covered by Sound On Sound and MusicRadar).
- **Recognition**, Government Medical Officers' Association, Sri Lanka, 2020, for a computer-vision system for remote COVID-19 patient vitals monitoring deployed across hospital wards.

Invited Talks, Demonstrations and Performances

Sep 2025	Demo, Notte della Ricerca, MUSE Science Museum, Trento
2025	Mixed-reality smart-instrument demo, MUSMET kickoff meeting, University of Trento
Oct 2024	Two live multisensory concerts with smart instruments, lights, MR and haptics, University of Trento
Sep 2024	Paper and demo, Audio Mostly 2024, University of Milan; demo, ICT Days, University of Trento
May 2024	Talk, *Real-time musical pattern detection*, McGill University, Montreal
Oct 2023	Demo, ISMIR 2023, Politecnico di Milano; demo, SoundMIT Synth Expo, Turin

Supervision

No publications co-authored with the supervised students to date.

- L. Collizoli, BSc thesis: *MIDI plugin for real-time musical pattern recognition*, University of Trento, Aug 2023.
- D. Battisti, BSc thesis: *Plugin Implementation and GUI for Real-Time Musical Pattern Detection*, University of Trento, Sep 2026.

Academic Service

- **Journal reviewer:** IEEE Access; Journal of the National Science Foundation of Sri Lanka.
- **Programme committee / reviewer:** IEEE International Symposium on the Internet of Sounds (IS²); International Conference on Immersive and 3D Audio (I3DA); Colloquio di Informatica Musicale (CIM).

Industry Research Experience

Independent Researcher Jan 2026 – present
Toronto, ON | open-source work and MUSMET external collaborator

- Released Nebula; researching foundation-model-based synthetic dataset curation for limited-label regimes; authored three publications on embedded real-time pattern detection and edge-to-server feature extraction.

Technical Consultant Aug 2020 – Feb 2021
Forestpin (Pvt) Ltd, Colombo, Sri Lanka

- ML and statistical pipelines for financial forensics, compliance monitoring and risk detection on large enterprise datasets (SQL, Python).

Research Engineer Jan 2015 – Jul 2020
MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka

- End-to-end computer-vision and IoT systems for manufacturing at scale, integrating camera and sensor streams into production; efficiency gains of up to 300% in deployed solutions.
- Care-label quality control (inspection time reduced by 99.5%); loom-side fabric inspection pilot; real-time IoT analytics across distributed factories; COVID-19 remote vitals device.
- Mentored engineers and introduced code review and CI/CD practices; ran group-wide computer-vision training.

Technical Skills and Languages

Languages: Python, C++17, C, SQL, MATLAB, JavaScript, C#. **ML:** TensorFlow/Keras, PyTorch, scikit-learn, RNN/LSTM/GRU, DTW, transfer learning. **Audio and signal processing:** MFCC, STFT, S-transform, librosa, SciPy, FFTW, JUCE, VST, OSC, MIDI. **Systems:** Raspberry Pi/ARM, Elk Audio OS, Docker, Airflow, AWS, Git, CI/CD. **Research:** experimental design, benchmark and dataset design, ablations, user studies, LaTeX, technical writing. **Languages:** English (native), Sinhala (mother tongue), Italian (B1), French (A2, learning).

References

Available on request. Contact details for three referees supplied in the application portal. Academic referees: Prof. Luca Turchet (University of Trento; PhD supervisor and postdoctoral principal investigator) and Prof. Marcelo M. Wanderley (McGill University; visiting-researcher host).